

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1504

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.*
(BL4)

Dave's Taxonomy: Articulation (Level 4)
Question Count: 5

Total Marks: 40

Duration :60 Minutes

Code of Conduct

I S Jefna Princy – 192572058 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Industry Problem Solving Task

1. **Retail Data Optimization Problem** - A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.
2. **Social Media Fraud Detection Problem** - A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.
3. **Government Data Platform Problem** A government agency is building a national data platform. It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.
4. **Banking Fraud Detection Problem**
A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data. Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement.

5. Agriculture Smart Farming Problem

A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.

Industry Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Industry Context	20%	Demonstrates strong understanding of real-world problem context	Good understanding	Basic understanding	Weak understanding
Application of Big Data Concepts	25%	Applies highly relevant big data ideas and tools	Mostly appropriate application	Partial application	Weak application
Solution Design	25%	Solution is practical, scalable, and well-reasoned	Reasonable solution	Basic solution with gaps	Unclear solution
Security / Risk Awareness	15%	Strong awareness of security and operational risks	Reasonable awareness	Limited awareness	Poor awareness
Clarity and Professionalism	15%	Well-structured and professional response	Generally clear	Moderately clear	Poorly presented

Total Marks Awarded:

INDUSTRY PROBLEM SOLVING TASK

1. Retail Data Optimization Problem

A retail chain operates both physical stores and an online platform, generating large volumes of sales, inventory, and customer behaviour data. The main requirements are real-time inventory tracking, demand forecasting, and personalized marketing. Therefore, a scalable big data architecture is required to integrate data from both offline and online sources.

Proposed Big Data Architecture

Data can be collected from Point-of-Sale (POS) systems, online transactions, inventory systems, websites, and customer applications. Real-time transaction data can be transferred using an event-streaming platform such as Apache Kafka. Historical and semi-structured data can be stored in distributed cloud storage or a data lake.

Apache Spark can be used for large-scale batch and real-time processing. A data warehouse can store processed business data for reporting and analysis. Machine learning models can analyse historical sales, seasonal patterns, customer behaviour, and product demand.

Analytics and Decision Making

Demand forecasting models can predict future product requirements and help prevent overstocking and stockouts. Customer behaviour can be analysed to generate personalized product recommendations and marketing offers.

The architecture should be scalable so that additional stores, customers, and transactions can be handled without major changes. Dashboards can provide managers with real-time information about sales, inventory levels, and predicted demand.

Security and Risks

Customer information and transaction data should be protected using encryption, authentication, role-based access control, and secure APIs. Data quality should also be monitored because inaccurate inventory or customer data can lead to incorrect business decisions.

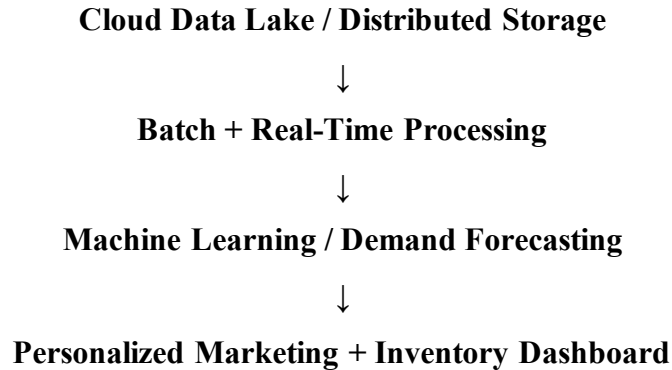
Overall flow:

Physical Stores / Online Platform / POS Systems



Data Integration & Ingestion Layer





2. Social Media Fraud Detection Problem

A social media platform processes billions of posts, comments, likes, and shares every day. Since fake accounts and misinformation must be detected quickly, the system requires scalable real-time or near-real-time processing.

Proposed Big Data Solution

User activities can be collected as continuous events using Apache Kafka or a similar streaming platform. The incoming data can be processed using Apache Spark Streaming or Apache Flink. The system can maintain user profiles containing information such as account activity, posting frequency, interaction patterns, and network behaviour. Distributed storage can maintain both historical and current user data.

Machine Learning and Fraud Detection

Machine learning can identify unusual behaviour associated with fake accounts. Classification models can use features such as unusual posting frequency, repeated content, abnormal interaction patterns, and suspicious account relationships.

For misinformation detection, Natural Language Processing (NLP) can analyse the text of posts and identify potentially misleading or suspicious content. Historical data can be used to train models, while streaming data can be used for real-time detection.

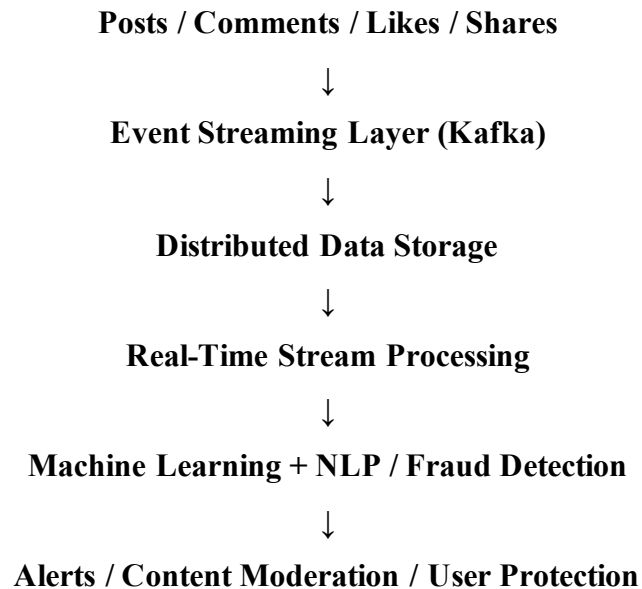
Scalability and Security

The streaming architecture can process events in parallel across multiple nodes, allowing the platform to handle billions of interactions. Data partitioning based on user ID, region, or time can improve processing performance.

Security should include authentication, access control, encryption, and protection of user information. False positives should also be considered because incorrectly blocking genuine users

can affect user experience.

Overall flow:



3. Government Data Platform Problem

A government agency is developing a national data platform that combines census, economic, and geospatial data from different departments. Since multiple agencies need to access shared information, interoperability, security, and data governance are major requirements.

Proposed Data Architecture

Data can be collected from different government departments through secure APIs and data integration pipelines. Structured information such as census and economic records can be stored in databases, while geospatial and other large datasets can be maintained in a distributed data lake.

An ETL/ELT pipeline can extract data from different agencies, transform it into standardized formats, and load it into the central platform. Metadata and data cataloguing can help departments understand the source, quality, and meaning of datasets.

Interoperability and Data Management

Standard APIs and common data formats should be used so that different government systems can exchange information. Data governance policies should define ownership, access permissions, data quality, retention, and usage.

Geospatial analytics can combine location information with census and economic data to support

planning, resource allocation, and public policy decisions.

Security and Risk Management

Government data may contain sensitive information, so strong security is required. Encryption, identity management, role-based access control, audit logs, and secure APIs can restrict unauthorized access.

Data quality checks should identify duplicate, missing, or inconsistent records before they are used for decision-making.

Overall flow:

Government Departments / Census / Economic / Geospatial Data



Data Integration & Secure APIs



Central Data Lake + Databases



ETL/ELT + Data Governance



Big Data Analytics / Geospatial Analytics



Policy Planning + Government Decision Making

4. Banking Fraud Detection Problem

A bank processes millions of financial transactions every minute and needs to detect fraudulent activity in real time. Since both historical and streaming transaction data are required, the solution should combine batch analytics with real-time stream processing.

Proposed Big Data Pipeline

Transaction events from ATMs, online banking, mobile applications, and card systems can be continuously collected through Apache Kafka. A stream-processing framework such as Apache Flink or Spark Streaming can analyse transactions immediately.

Historical transaction data can be stored in a data lake or warehouse. Batch processing can identify long-term customer behaviour and previous fraud patterns, while real-time processing can identify suspicious transactions as they occur.

Algorithms and Analytics

Machine learning models such as Random Forest, Logistic Regression, or Gradient Boosting can classify transactions as normal or suspicious. Anomaly detection can also identify unusual transaction amounts, locations, frequencies, or spending patterns.

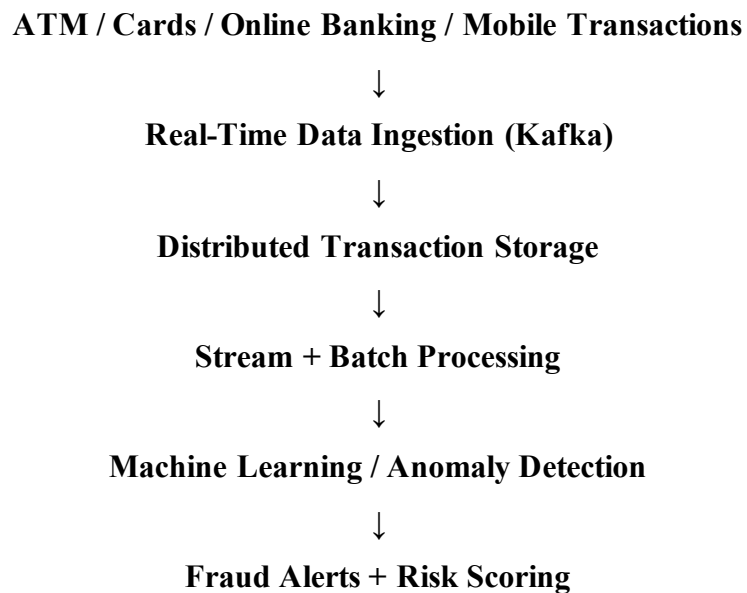
A transaction risk score can be generated by combining multiple indicators. High-risk transactions can be flagged for additional verification or investigation.

Security and Reliability

Banking systems require very high security and accuracy. Data should be encrypted during storage and transmission, with strict authentication and role-based access control.

The system should also be fault tolerant because failure of fraud detection services could create financial and security risks. Distributed processing and replicated data can improve availability.

Overall flow:



5. Agriculture Smart Farming Problem

The smart farming system collects soil, weather, and crop data using sensors and drones. Since the information comes from multiple distributed sources and is generated continuously, a big data solution can help farmers make accurate and timely decisions.

Proposed Big Data Architecture

IoT sensors can continuously collect soil moisture, temperature, humidity, nutrient levels, and other field information. Drones can provide crop images that can be analysed to identify crop

health and possible disease.

Sensor data can be transmitted through an IoT gateway and stored in cloud-based or distributed storage. Streaming technologies can process real-time sensor readings, while historical data can be analysed using batch processing.

Analytics and Decision Making

Machine learning models can analyse soil, weather, and historical crop information to predict crop conditions and yield. Image analytics can identify unhealthy crop areas from drone images. The system can provide recommendations for irrigation, fertilizer usage, crop monitoring, and resource allocation. For example, if soil moisture falls below a required level, the system can generate an irrigation recommendation.

Scalability, Security and Benefits

Cloud-based storage allows the system to handle data from multiple farms and fields. Data processing can be distributed so that large volumes of sensor and image data can be analysed efficiently.

IoT devices should use authentication and secure communication to prevent unauthorized access.

Farmers should have controlled access to their farm data.

The solution supports precision agriculture by reducing unnecessary resource usage and helping farmers make data-driven decisions.

Overall flow:

